Reproducing Transfer-Learning and Snapshot-Ensemble Diabetic Retinopathy Grading on APTOS 2019: Stage 2 (Base Paper & Reproducibility Check) and Stage 3 (Baseline Reproduction)

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CV Research-Reproduction Project — Diabetic Retinopathy Grading

Abstract—We select Chilukoti et al. (2024), which reports a quadratic weighted kappa (QWK) of 0.954 on APTOS 2019 with an ImageNet-initialised EfficientNet-B3 and 0.967 with EyePACS pre-training and snapshot ensembling, as our base paper, document its reproducibility, and re-implement it from the text because no code is public. Under hardware limits (Apple M2, 8 GB), we reproduce the method at 224 px for 30 epochs. With the paper's hyper-parameters we obtain test QWK 0.827 (macro-F1 0.473), Severe NPDR is never predicted, and the paper's snapshot ensemble lowers QWK to 0.788 because training is unstable. Changing only the learning rate to 10^{-4} stabilises training and yields QWK 0.887 (macro-F1 0.630, Severe/PDR recall 0.31/0.55), closing about half of the gap. We also find 123 groups of byte-identical images in APTOS (30 with conflicting grades), whose leakage into the test set has no measurable effect. No clinical claims are made.

I. Stage 2: Base Paper and Reproducibility Check

A. Selected base paper

We reproduce Chilukoti et al., "A reliable diabetic retinopathy grading via transfer learning and ensemble learning with quadratic weighted kappa metric," BMC Medical Informatics and Decision Making, 24:37, 2024 [1]. The paper fine-tunes an ImageNet-pretrained EfficientNet-B3 with a three-layer classifier head, preprocesses fundus images with CLAHE followed by Gaussian blur, and ensembles predictions from weights saved at several epochs of a single training run ("snapshot" ensembling). It reports QWK as the primary metric: 0.954 on APTOS 2019 for a single model trained from ImageNet weights, and 0.967 when the model is first trained on EyePACS and the snapshot ensemble is used.

B. Why this paper

It matches the direction set by our literature review: it treats QWK (an ordinal, imbalance-aware agreement metric) as primary, uses a transfer-learning CNN that fits free-tier hardware, evaluates on the public five-class APTOS 2019 dataset, and leaves the two gaps we identified untouched: no imbalance-aware loss and no ordinal-aware training objective. That makes it a clean base for a single-variable Stage 4 hypothesis.

C. Reproducibility checklist

TABLE I
Reproducibility checklist for the base paper.

Item	Our answer
Public code?	No. No repository is linked and none was found on GitHub; we re-implemented the method from the paper text (Sec. II).
Dataset public & size?	Yes. APTOS 2019 Blindness Detection (Kaggle), 3,662 labelled fundus images, ≈8.9 GB at full resolution; obtained from a full-resolution Hugging Face mirror (sngsfydy/aptos). Class counts 1805/370/999/193/295.
Fits free GPU?	Yes on Colab/Kaggle T4 at 300×300. Locally (Apple M2, 8 GB unified memory) B3 at 300 px swaps heavily even at batch 8, so the local run uses 224 px (see deviations).
Last commit / usable?	Not applicable (no code). Our implementation: PyTorch 2.14, timm 1.0.30.
Metric defined?	Yes. QWK (primary), plus accuracy, precision, recall, F1, AUC-ROC. We additionally report macro-F1, per-class precision/recall/F1, macro one-vs-rest AUC and the confusion matrix.
Split defined?	Partially. A 70/15/15 train/val/test split is stated for EyePACS only; not stated for APTOS. We use a stratified 70/15/15 split with a fixed seed.
Hyper-parameters?	Partially. Adam, lr 10^{-3}, weight decay 10^{-4}, gradient clipping 0.1, 60 epochs (strategy 1). Batch size, APTOS epoch count, loss function, CLAHE parameters and blur kernel are not stated.

D. Under-specified details and our choices

- Loss: not stated; softmax output implies cross-entropy, which we use.
- Gradient clipping "0.1": interpreted as global L2-norm clipping at 0.1.
- CLAHE / blur: clip limit 2.0, 8×8 tiles on the LAB luminance channel, then a 5×5 Gaussian blur, applied identically to train, validation and test images.

- Resize: plain square resize (no border cropping), as the paper mentions only resizing.
- Augmentation: none beyond CLAHE and blur, which the paper calls "augmentation" but which act as deterministic pre-processing.
- Ensemble: strategy 2 takes the majority vote over predictions from 10 evenly spaced snapshots; strategy 1 averages the 30- and 60-epoch (here: mid- and final-epoch) snapshots.
- Reported model: the paper does not say whether the final-epoch or best-validation model is evaluated; we report both.

E. Red flags noted before reproduction

Several reported numbers are internally inconsistent, which we flag as context for the reproduction gap: (i) in Table 3 of [1], precision, recall and QWK of the B3 ensemble are all exactly 0.867, and models predicting only class 0 report a macro AUC of 0.196, below the 0.5 that a constant predictor yields; (ii) in Table 6, accuracy, precision, recall and F1 agree to within 0.01 on an imbalanced dataset; (iii) the APTOS split is not described, so overlap between near-duplicate images in train and test (a known APTOS issue) cannot be ruled out. We audit duplicates ourselves in Sec. II-B.

II. Stage 3: Reproduction Protocol

A. Data

APTOS 2019 [2] contains 3,662 labelled colour fundus photographs from Aravind Eye Hospital graded 0–4 (No DR 1805, Mild 370, Moderate 999, Severe 193, PDR 295). Images were stored at a 512 px longest side (aspect ratio preserved) and resized at load time. We use a stratified 70/15/15 split with seed 42 (2,562/550/550 images).

B. Duplicate audit

Hashing the raw bytes revealed 123 groups of byte-identical images (128 redundant rows); in 30 of these groups the copies carry different grades (e.g. the same photograph labelled Moderate and PDR). Under a naive random split, 29 of the 550 test images (5.3%) have an identical copy in the training set, and 5 of them with a conflicting grade. We train on the faithful split (all 3,662 rows, as the paper presumably did) and quantify the effect of this leakage by re-scoring each model on the 521 test images that have no copy in the training set (Sec. III-D). A de-duplicated protocol (conflicting groups removed, one copy per group: 3,504 images) is implemented for Stage 5.

C. Model and training

Following [1]: an ImageNet-pretrained EfficientNet-B3 [3] (timm) with global average pooling (1,536-d) feeds a head of FC(512), Dropout(0.5), ReLU, FC(512), Dropout(0.25), ReLU and FC(5). The whole network is fine-tuned with Adam (lr 10^{-3}, weight decay 10^{-4}), global gradient-norm clipping at 0.1 and cross-entropy loss. Inputs are normalised with ImageNet statistics.

Strategy 2 (headline): CLAHE [4] on the LAB luminance channel (clip 2.0, 8×8 tiles), 5×5 Gaussian blur, square resize. Predictions are taken from 10 evenly spaced epochs and combined by majority vote. Strategy 1: plain resize to 150 px; predictions from the mid-point and final epochs are averaged.

D. Deviations forced by the hardware

All runs were trained locally on an Apple M2 with 8 GB unified memory (PyTorch MPS backend). EfficientNet-B3 at 300×300 exceeded the memory budget even at batch size 8 (training steps took 25–45 s because of swapping), whereas 224×224 at batch 8 used 2.8 GB and ran at 12.4 images/s. We therefore use 224 px instead of 300 px for strategy 2, batch size 8 (16 for the 150 px strategy 1), and 30 instead of 60 epochs (snapshots every 3 epochs instead of every 6). One run takes ≈2 h. The exact protocol (300 px, 60 epochs, batch 16) is a single flag change and is provided for a free CUDA GPU (Colab/Kaggle). Training is seeded and was bit-for-bit repeatable on MPS for the 13 epochs we could compare across two launches; the headline run was interrupted once (disk full while writing a checkpoint) and resumed from its epoch-13 checkpoint, after which data-loader shuffling differs from an uninterrupted run. The EyePACS-pretraining variant behind the paper's 0.967 was not attempted: EyePACS is 35,126 images (≈10× APTOS), which is beyond the local compute budget, so our target is the paper's ImageNet-initialised result (QWK 0.954).

E. Metrics

Following the course rules and our literature review, we report QWK [5] (primary), accuracy, macro-F1, macro one-vs-rest AUC, per-class precision/recall/F1, "minority recall" (mean recall of Severe and PDR), the number of test predictions off by two or more grades, and confusion matrices. We report the final-epoch model, the best-validation-QWK model, and both snapshot-ensemble variants.

III. Stage 3: Results

A. Faithful reproduction

Table II summarises the test results. With the paper's hyper-parameters, the single model reaches QWK 0.827 (accuracy 0.771, macro-F1 0.473), against 0.954 reported in [1]: a gap of roughly 0.13 QWK. In this run the best-validation epoch was the final epoch, so both rows coincide. The 10-snapshot majority vote, the paper's central contribution, lowers QWK to 0.788.

Training was unstable (Fig. 1): validation QWK dropped sharply at epochs 2, 5, 10, 17 and 23 (to −0.006 at epoch 5, when the model predicted almost only one grade) and recovered within one or two epochs, while training loss was still 0.61 after 30 epochs. Test QWK of the individual snapshots (epochs 3, 6, …, 30) was 0.645, 0.757, 0.772, 0.789, 0.754, 0.770, 0.748, 0.761, 0.750 and 0.827: only the

TABLE II
Reproduced test-set results on APTOS 2019 (550 test images, faithful split). Paper: QWK 0.954 (single model, ImageNet init) and 0.967 (EyePACS init + snapshot ensemble).

Configuration	Prediction	QWK	Acc.	Macro-F1	Macro AUC	Minority recall	$\lvert\Delta\rvert \geq 2$ errors
Strategy 2, lr 10^{-3} (paper)	final epoch	0.827	0.771	0.473	0.897	0.148	48
	best val QWK	0.827	0.771	0.473	0.897	0.148	48
	snapshot vote	0.788	0.767	0.412	0.913	0.000	50
	snapshot mean	0.791	0.769	0.416	0.913	0.000	49
Strategy 2, lr 10^{-4} (diagnostic)	final epoch	0.862	0.798	0.603	0.897	0.370	30
	best val QWK	0.887	0.815	0.630	0.894	0.428	24
	snapshot vote	0.887	0.822	0.613	0.930	0.347	27
	snapshot mean	0.887	0.820	0.611	0.930	0.347	27

TABLE III
Per-class test metrics, strategy 2 (faithful split).

Grade	Model	Prec.	Recall	F1	n
No DR	best val	0.96	0.96	0.96	271
Mild	best val	0.54	0.23	0.33	56
Moderate	best val	0.62	0.93	0.75	150
Severe	best val	0.00	0.00	0.00	29
PDR	best val	0.39	0.30	0.34	44
No DR	snapshot vote	0.95	0.97	0.96	271
Mild	snapshot vote	0.52	0.29	0.37	56
Moderate	snapshot vote	0.59	0.96	0.73	150
Severe	snapshot vote	0.00	0.00	0.00	29
PDR	snapshot vote	0.00	0.00	0.00	44

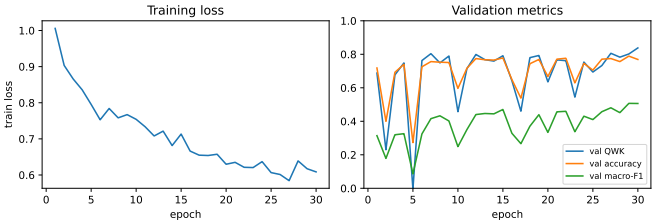


Fig. 1. Training loss and validation metrics, strategy 2 with the paper's lr 10^{-3}.

last snapshot is strong, so a majority vote is dominated by weaker models. Snapshot ensembling assumes comparably good, diverse members [6], which an unstable trajectory without a cyclic learning-rate schedule does not provide.

B. Per-grade behaviour

QWK and accuracy hide a severe minority-grade failure (Table III, Fig. 2). No DR (recall 0.96) and Moderate (0.93) are learned well, but Severe NPDR is never predicted (recall 0.00 in every snapshot), PDR recall is 0.30 for the single model and 0.00 for the vote, and Mild recall is 0.23. Severe cases are split between Moderate (15/29) and PDR (13/29), and 27 of 44 PDR eyes are called Moderate. The errors are mostly adjacent-grade, which is why QWK stays near 0.83 even though two of five grades are essentially unrecognised; this is exactly the "accuracy (and even QWK) can be misleading" gap identified in our literature review.

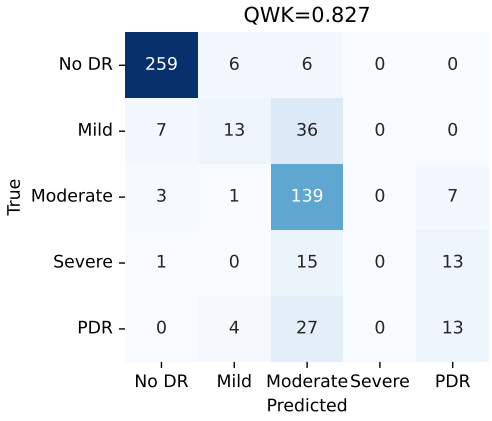


Fig. 2. Test confusion matrix of the reproduced single model (paper hyper-parameters).

C. Diagnostic: learning rate 10^{-4}

To test whether optimisation explains the gap, we repeated strategy 2 with a single change, lr $10^{-3} \rightarrow 10^{-4}$ (same split, seed, resolution, epochs and snapshots). Training became stable (no collapses; training loss 0.27–0.32 after 20 epochs versus 0.61 at the end of the paper-setting run) and every metric improved (Table II): QWK 0.887 for the best-validation model (epoch 11) and 0.887 for the 10-snapshot vote, accuracy 0.822, macro-F1 0.630. Minority recall (mean of Severe and PDR) rose from 0.148 to 0.428: Severe 0.00→0.31 and PDR 0.30→0.55 for the single model, and predictions off by two or more grades halved (48→24). With stable training the snapshots are uniformly good (test QWK 0.834–0.875), and the vote now matches the best single model, consistent with the snapshot-ensemble assumption that members are of comparable quality.

This closes about half of the gap to 0.954 (remaining ≈0.07 QWK) and indicates that the paper's stated learning rate, at least at our batch size, is the main cause of the faithful run's poor result. Severe NPDR remains the weakest grade (single-model recall 0.31; 0.10 for the vote), and Mild is still mostly confused with Moderate (24/56). Leakage again has no measurable effect (QWK 0.887 with

and without the 29 duplicated test images). This run is the baseline we will carry into Stage 4.

D. Reproduction-gap analysis

- Duplicate leakage does not explain it. Excluding the 29 test images with an identical copy in training changes QWK from 0.827 to 0.828. The leaked images are actually predicted worse (65.5% accuracy), because several copies carry conflicting grades.
- Optimisation (learning rate). Full fine-tuning with Adam at 10^{-3}, gradient clipping at 0.1 and batch size 8 is unstable (repeated collapses). The paper's own Table 3, where VGG and ResNet models predict only class 0 under the same settings, is consistent with this. See the diagnostic run below.
- Resolution and schedule. We trained at 224 px for 30 epochs instead of 300 px for 60; small lesions (microaneurysms) that separate Mild from No DR and Severe from Moderate are less visible at lower resolution.
- Unreported details. Split, batch size, loss, CLAHE parameters, and whether the reported model is the final or the best-validation one are not stated; each can move QWK by a few points on a 550-image test set.
- Reported numbers. The internal inconsistencies noted in Sec. 2 (e.g. identical precision, recall and QWK; macro AUC below 0.5) mean the reported 0.954 may itself not be reliable.

E. Failure cases

Fig. 3 shows the most confidently wrong test predictions. Most are PDR eyes predicted as Mild or Moderate, including one with a large pre-retinal haemorrhage; several are dim, low-contrast or partially cropped photographs, where neovascularisation is hard to see at 224 px.

IV. Limitations

Single dataset (APTOS 2019) and a single seed; no external validation; 224 px / 30-epoch protocol instead of 300 px / 60 epochs because of local hardware; the EyePACS-pretrained configuration (0.967) was not reproduced. The 550-image test set contains only 29 Severe and 44 PDR images, so per-grade recall estimates are noisy. This is a reproduction study; no clinical claims are made.

V. Next Step (Stage 4 preview)

The reproduction shows that the paper's pipeline fails on exactly the grades that matter most clinically. Our Stage 4 hypothesis will therefore change one variable, the training loss, from cross-entropy to a class-balanced, ordinal-aware objective, on top of the stable lr-10^{-4} baseline (QWK 0.887, Severe recall 0.31), and will be stated with a predicted direction and magnitude before any code is run.

References

[1] S. V. Chilukoti, L. Shan, V. S. Tida, A. S. Maida, and X. Hei, "A reliable diabetic retinopathy grading via transfer learning and ensemble learning with quadratic weighted kappa metric," BMC Medical Informatics and Decision Making, vol. 24, p. 37, 2024.

[2] Karthik, Maggie, and S. Dane, "APTOS 2019 blindness detection," Kaggle competition, https://kaggle.com/competitions/aptos2019-blindness-detection, 2019.

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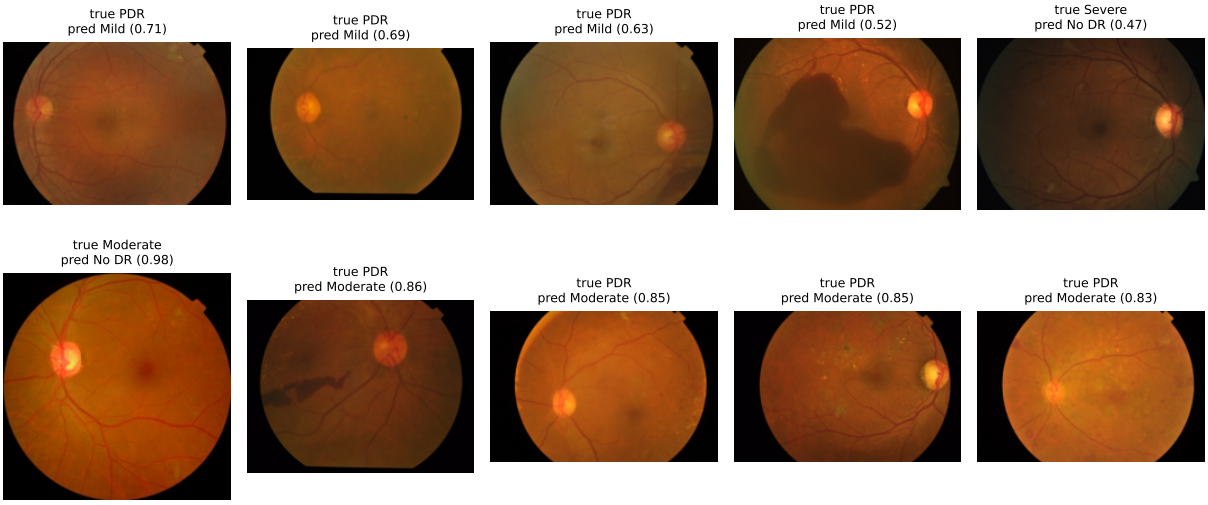


Fig. 3. Ten largest-error test predictions of the reproduced single model (true grade, predicted grade, confidence).